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# Mahmud Zaman Bali

Luton, United Kingdom
mzbali.com

GitHub: github.com/mzbali
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## Summary

Software engineer building .NET and React web applications, from server-rendered front ends to the pipelines and Kubernetes deployments that ship them. Also builds native macOS tools in Swift.

## Experience

**Law Business Research** — London, United Kingdom
*Web Developer* — *Apr 2025 – Present*
*Junior Web Developer* — *May 2024 – Apr 2025*

- Build and maintain .NET (C#) APIs and server-rendered React front ends.
- Build MCP tools that connect internal systems to AI assistants.
- Ship services to Kubernetes with Helm charts through Azure DevOps pipelines.
- Mentor a junior developer; write technical specifications, plan and estimate work.

**Teaching Assistant (Computer Science)** — Dhaka, Bangladesh
*American International University-Bangladesh* — *2020 – 2021*

- Developed lesson materials for CS courses, emphasising programming concepts and problem-solving.
- Provided focused tutoring to students, aiding in programming challenges and code debugging.

## Education

**Queen Mary University of London** — London, United Kingdom
*MSc, Artificial Intelligence* — *2022 – 2023*

- **1st** class **Distinction**.

**American International University-Bangladesh** — Dhaka, Bangladesh
*BSc, Computer Science and Engineering* — *2017 – 2021*

- **GPA:** 3.93/4.00
- **Achievements:** Dean's List Award.

## Technical Skills

**Web Development:** .NET Core, React, Next.js, C#, TypeScript, JavaScript
**DevOps:** Kubernetes, Helm, Azure DevOps, Docker, GitHub Actions, Git, Bash
**Databases:** PostgreSQL, Redis
**Other:** Swift, MCP, Rust, Linux, macOS
**Languages:** English (fluent), Bengali (native)

## Projects

**ClipVee** | *Swift, SwiftUI, AppKit, Core Data, Sparkle*

- Windows-style clipboard history for macOS: a card-based, keyboard-first fork of Maccy.
- Ships as a signed DMG and Homebrew cask, with in-app updates through Sparkle.

**BugBlitz** | *.NET Core, Next.js, PostgreSQL, Docker, Azure*

- Developed using .NET Core 8 and Next.js by adhering to SOLID guidelines.
- Decreased code complexity by 20% by delegating authentication to an OAuth2 microservice.
- Applied caching in Next.js, substantially cutting back on database queries and boosting efficiency.

**Storetail** | *.NET Core, React, Redux, PostgreSQL, JWT, Stripe*

- Leveraged Extension Pattern with SOLID principles for enhanced code organisation and readability.
- Formulated a CI/CD pipeline using GitHub Actions, enabling an efficient development-to-deployment transition.